

No Path: Cross-Domain Derivation by an Outsider Using Commodity Tools

A Case Study in Institutional Exclusion of Valid Work

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1. What Was Done

Two integers from gauge theory — 11 and 13 — were passed through standard cosmological equations to predict how much deuterium, helium, and lithium the universe produced in its first minutes after the Big Bang. The predictions were compared against Planck satellite measurements and precision spectroscopy of distant gas clouds. The results:

Deuterium abundance: predicted 2.531×10^{-5} , measured 2.527×10^{-5} . Three-digit agreement. Miss 0.14%. Deviation from measurement: 0.12 standard deviations.

Helium-4 mass fraction: predicted 0.2486, measured 0.2449. Within one standard deviation. Miss 1.53%. Deviation from measurement: 0.94 standard deviations.

Baryon density parameter: predicted 0.049036, measured 0.0490. Miss 727 parts per million.

Lithium-7 abundance: predicted approximately $3 \times$ higher than observed. This is the known lithium problem — standard Big Bang Nucleosynthesis predicts too much lithium regardless of input source. The chain reproduces the discrepancy at the correct ratio, confirming it runs the same physics as every other BBN calculation.

The derivation chain crosses three physics domains. It starts in particle physics — gauge integers from the Standard Model. It crosses into cosmology — baryon density, photon

density, baryon-to-photon ratio. It lands in nuclear physics — primordial element abundances. The chain spans over twenty-five orders of magnitude in energy scale.

No one in the institutional physics community has published this chain. Not because the equations are secret. Not because the values are unavailable. Not because the computation is beyond current capability. The equations are in textbooks. The values are in reference compilations. The computation is trivial. The chain has not been published because it crosses three departmental boundaries and no department owns it.

The chain was produced by a person with no physics degree, no institutional affiliation, and no publication history in physics journals. The tools were a laptop, Python, and a large language model for search and derivational assistance. The time investment was three months. The physics background was high school level.

2. The Tools

The computation uses four tools. Each is commodity technology available to anyone.

Python. The programming language. Free, open-source, installed on every operating system. The built-in `Fraction` class provides exact rational arithmetic — numerator and denominator stored as arbitrary-precision integers, every operation exact, no rounding, no accumulated error.

mpmath. A Python library for arbitrary-precision floating-point arithmetic. Used where intermediate computations require high precision before conversion to exact rationals. Free, open-source, installable with one command.

A large language model. Provides search and derivational assistance. Knows the Standard Model equations. Knows the BBN fitting formulae. Knows the CODATA values. Knows how to implement exact rational arithmetic in Python. Any commercial or open-source LLM with training data that includes physics textbooks provides equivalent capability.

A JSON-based value store and experiment runner. Custom infrastructure — approximately 2,000 lines of Python — that stores named values with provenance, loads declared inputs for experiments, executes derivations, evaluates pre-registered comparisons, and writes versioned results. Standard data engineering. No physics knowledge in the infrastructure. The runner does not know what a baryon is.

No Mathematica. No MATLAB. No lattice QCD code. No Monte Carlo simulation. No specialized physics software. The computation runs on any machine with Python installed.

3. The Arithmetic

Physics uses floating-point arithmetic for nearly all computation. A 64-bit IEEE 754 float has 52 bits of mantissa — approximately 15 to 16 significant digits. Every arithmetic operation can introduce a rounding error in the last represented bit. Over a derivation chain of hundreds of operations crossing three physics domains, these errors accumulate. The accumulated error is untracked. It is absorbed into reported uncertainties and attributed to physics rather than to arithmetic.

Exact rational arithmetic does not round. The fraction 22/13 is stored as numerator 22, denominator 13. Multiplied by 3/7, it produces 66/91, which reduces to 6/7. Every digit is exact. The chain can be a thousand steps long and the last digit is as precise as the first. When the final result disagrees with a measurement, the disagreement is in the physics or in the measurement. Zero percent of the disagreement is in the arithmetic.

The physics community already has exact rational values for much of its foundation. The 2019 SI redefinition made several fundamental constants exact by definition:

speed of light:	$299792458 / 1$	m/s
Boltzmann constant:	$1380649 / 10^{29}$	J/K
Planck constant:	$132521403 / 2 \times 10^{41}$	J·s
Avogadro's number:	$6022140760000000000000000 / 1$	
elementary charge:	$801088317 / 5 \times 10^{27}$	C

These are exact. Not approximate. Defined as exact integers or exact fractions by international agreement. Physics already committed to exact rational representation for its most fundamental quantities. It then converts them to floating-point before computing with them.

The Standard Model beta function coefficients — which describe how coupling strengths change with energy scale — are exact fractions derived from the gauge group structure:

$b_1(U(1)) = 41/10$	$b_1(\text{modified}) = 25/6$
$b_2(SU(2)) = -19/6$	$b_2(\text{modified}) = -13/6$
$b_3(SU(3)) = -7$	$b_3(\text{modified}) = -20/3$

Exact. From group theory. Known for decades. Converted to floats before use.

The QED perturbative coefficients — the terms in the most precise prediction in science — include exact rational components:

Schwinger term (a1): 1/2
a2 rational piece: 197/144
a2 $\zeta(3)$ coefficient: 3/4
a2 π^2 coefficient: 1/12
a3 rational term: 28259/5184

Exact. Published in analytical form. Converted to floats before use.

The Casimir ratios from group theory: 2, 3, 4/3, 3/4. The representation theory factors: 1/15, 1/3, 1/6, 2/3. The generation democracy sum: 4/3 for each gauge group. All exact. All converted to floats before use.

The only quantities that are not naturally rational are the transcendental numbers — π , Euler's number e , the Riemann zeta values $\zeta(3)$, $\zeta(5)$, natural logarithms, and elliptic integrals. These have infinite decimal expansions. No finite representation can capture them exactly.

The engineering solution: choose a common denominator large enough that the rational approximation exceeds any instrument's measurement precision. The denominator 2^{335} provides over 100 decimal digits of precision. Every transcendental is computed to more than 100 digits and stored as an integer numerator over 2^{335} .

For example, π stored as an exact fraction with the 2^{335} denominator:

numerator: 109943212936596175505913298521620533097335915961
174408408712911656578469374718859347550214371967
627157
denominator: 34996011596528190789960035633881941845650710894
291398982812329702559247987190014771576210832368
861184

The approximation error is below 10^{-100} . No instrument on Earth measures to 100 digits. The representation is exact for all practical purposes. The approximation error is quantified and bounded — unlike floating-point, where accumulated error is untracked.

The value store contains over sixty transcendental and special function values stored in this format: π , π^2 , e , $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt{7}$, the golden ratio, $\ln(2)$, $\ln(3)$, $\ln(5)$, $\zeta(2)$ through $\zeta(9)$, the Catalan constant, polylogarithms Li_4 through Li_7 at 1/2, and elliptic integrals K and E at 1/4, 1/2, and 3/4. Each stored with full provenance, each queryable, each exact to over 100 digits.

4. The Code

The derivation functions are Python. They implement textbook equations. A first-year computer science student could write them. A first-year computer science student could read them. They are trivial. They solve a problem the field has never solved.

The baryon density from integers

The dark matter to baryon ratio in the universe has a prefactor of $22 \pi / 13$. This comes from gauge integers 11 and 13 through the beta function structure. The baryon density is derived by dividing the measured dark matter density by this ratio:

```
def bridge_omega_b_from_integers_v0(value_dicts):
    vm = _value_map(value_dicts)
    pi_val = _mpf_val(vm, "geom_pi_v0")
    omega_dm = _mpf_val(vm, "cosmo_omega_dm_planck_v0")
    prefactor_ratio = mpf("22") / mpf("13")
    dm_baryon_ratio = prefactor_ratio * pi_val
    omega_b = omega_dm / dm_baryon_ratio
    omega_b_measured = _mpf_val(vm, "cosmo_omega_b_planck_v0")
    miss = abs(omega_b - omega_b_measured) / omega_b_measured * 100
    return {
        "key": "bridge_omega_b_from_integers_v0",
        "outputs": {
            "result_omega_b_derived_v0": _approx(omega_b),
            "result_omega_b_measured_v0": _approx(omega_b_measured),
            "result_omega_b_miss_pct_v0": _approx(miss),
            "result_dm_baryon_ratio_used_v0": _approx(dm_baryon_ratio),
```

```

    },
}

```

Two integers. One ratio. One division. Compare against Planck. The function loads every input from the value store by key. No hardcoded numbers. The result: 0.049036 versus Planck's 0.0490. Miss 727 parts per million. That is the entire function. It takes the field's own dark matter measurement, divides by a ratio derived from gauge integers, and lands within 0.07% of the Planck satellite measurement of baryon density.

The baryon-to-photon ratio

The baryon-to-photon ratio η connects cosmological density parameters to nuclear physics. It is computed from the baryon density, the CMB photon number density, and the critical density of the universe:

```

def bridge_eta_from_omega_b_v0(value_dicts):
    vm = _value_map(value_dicts)

    omega_b = mpf(str(_get(vm, "result_omega_b_derived_v0")))

    T_cmb = _mpf_val(vm, "cosmo_t_cmb_v0")

    rho_crit = _mpf_val(vm, "cosmo_rho_crit_v0")

    k_B = _mpf_val(vm, "si_boltzmann_constant_v0")

    c = _mpf_val(vm, "si_speed_of_light_v0")

    hbar = _mpf_val(vm, "si_planck_constant_v0") / (2 * pi)

    m_p = _mpf_val(vm, "mass_proton_v0") * mpf("1e6") * e_charge / (c * c)

    zeta3 = _mpf_val(vm, "geom_zeta3_v0")

    n_gamma = 2 * zeta3 / (pi * pi) * (k_B * T_cmb / (hbar * c)) ** 3

    n_b = omega_b * rho_crit / m_p

    eta = n_b / n_gamma

    return {
        "key": "bridge_eta_from_omega_b_v0",

```

```

    "outputs": {

        "result_eta_derived_v0": _approx(eta),

        "result_eta10_derived_v0": _approx(eta * mpf("1e10")),

        "result_n_gamma_computed_v0": _approx(n_gamma),

        "result_rho_crit_computed_v0": _approx(rho_crit_si),

        ...

    },

}

```

Standard cosmological formula. Photon number density from the CMB temperature using the Bose-Einstein distribution. Baryon number density from the baryon density parameter and the critical density. Ratio of the two. Every input loaded by key from the value store. The function is the equation translated into Python syntax. Nothing more.

The primordial deuterium abundance

```

def bridge_dh_from_eta_v0(value_dicts):

    vm = _value_map(value_dicts)

    eta10 = mpf(str(_get(vm, "result_eta10_derived_v0")))

    dh_a = _mpf_val(vm, "bbn_dh_a_coeff_v0")

    dh_b = _mpf_val(vm, "bbn_dh_b_coeff_v0")

    dh_x1e5 = dh_a + dh_b * (eta10 - mpf("6"))

    dh_derived = dh_x1e5 * mpf("1e-5")

    dh_measured = _mpf_val(vm, "cosmo_dh_measured_v0")

    dh_measured_unc = _mpf_val(vm, "cosmo_dh_measured_unc_v0")

    miss = abs(dh_derived - dh_measured) / dh_measured * 100

    dh_sigma = abs(dh_derived - dh_measured) / dh_measured_unc

```

```

return {

    "key": "bridge_dh_from_eta_v0",

    "outputs": {

        "result_dh_derived_v0": _approx(dh_derived),

        "result_dh_measured_v0": _approx(dh_measured),

        "result_dh_miss_pct_v0": _approx(miss),

        "result_dh_sigma_v0": _approx(dh_sigma),

    },

}

```

$D/H (\times 10^5) = a + b \times (\eta_{10} - 6)$. That is the Pitrou et al. 2018 fitting formula. One line of arithmetic. The coefficients $a = 2.57$, $b = -0.44$ are loaded from the value store, sourced to Pitrou et al. The derived η_{10} is loaded from the previous step's output. Compare against the Cooke et al. 2018 measurement. Compute miss. Compute sigma.

Result: predicted 2.531×10^{-5} , measured 2.527×10^{-5} . Three-digit match. Miss 0.14%. Sigma 0.12. The integer-derived prediction of how much deuterium the universe produced in its first three minutes matches precision spectroscopy of gas clouds at redshift 3 to three significant figures.

The function is nine lines of arithmetic and ten lines of output formatting. A first-year CS student could write it. Not could read it. Could write it. Given the equation and the value store keys, the function writes itself. It is a mechanical translation of a textbook formula into Python syntax. There is nothing else to it.

The primordial helium abundance

```

def bridge_yp_from_eta_v0(value_dicts):

    vm = _value_map(value_dicts)

    eta10 = mpf(str(_get(vm, "result_eta10_derived_v0")))

    yp_a = _mpf_val(vm, "bbn_yp_a_coeff_v0")

    yp_b = _mpf_val(vm, "bbn_yp_b_coeff_v0")

```



```

yp_derived = yp_a + yp_b * (eta10 - mpf("6"))
yp_measured = _mpf_val(vm, "cosmo_yp_measured_v0")
yp_measured_unc = mpf("0.0040")
miss = abs(yp_derived - yp_measured) / yp_measured * 100
yp_sigma = abs(yp_derived - yp_measured) / yp_measured_unc
return {
    "key": "bridge_yp_from_eta_v0",
    "outputs": {
        "result_yp_derived_v0": _approx(yp_derived),
        "result_yp_measured_v0": _approx(yp_measured),
        "result_yp_miss_pct_v0": _approx(miss),
        "result_yp_sigma_v0": _approx(yp_sigma),
    },
}

```

Same formula. Same structure. Different coefficients — $a = 0.2485$, $b = 0.0016$, from Pitrou et al. Same η_{10} from the chain. Compare against Aver et al. 2015.

Result: predicted 0.2486, measured 0.2449. Within one sigma. The digit test at three digits fails — the report prints FAIL. The sigma test passes — the report prints PASS. Both verdicts printed mechanically by the runner from pre-registered comparison criteria.

The lithium-7 abundance and the lithium problem

```

def bridge_li7_from_eta_v0(value_dicts):
    vm = _value_map(value_dicts)
    eta10 = mpf(str(_get(vm, "result_eta10_derived_v0")))
    li7_a = _mpf_val(vm, "bbn_li7_a_coeff_v0")
    li7_b = _mpf_val(vm, "bbn_li7_b_coeff_v0")

```

```

li7_x1e10 = li7_a + li7_b * (eta10 - mpf("6"))
li7_derived = li7_x1e10 * mpf("1e-10")
li7_measured = _mpf_val(vm, "cosmo_li7_measured_v0")
li7_measured_unc = _mpf_val(vm, "cosmo_li7_measured_unc_v0")
miss = abs(li7_derived - li7_measured) / li7_measured * 100
li7_sigma = abs(li7_derived - li7_measured) / li7_measured_unc

return {
    "key": "bridge_li7_from_eta_v0",
    "outputs": {
        "result_li7_derived_v0": _approx(li7_derived),
        "result_li7_measured_v0": _approx(li7_measured),
        "result_li7_miss_pct_v0": _approx(miss),
        "result_li7_sigma_v0": _approx(li7_sigma),
    },
}

```

Same formula. Same structure. Coefficients $a = 4.68$, $b = 0.67$, from Pitrou et al. Same η_{10} . The result: predicted lithium abundance approximately three times higher than observed. This is the cosmological lithium problem. It has been open for decades. The chain reproduces it at the correct ratio.

This is a validation. The chain uses the same equations as every other BBN calculation. It gets deuterium right. It gets helium right. It gets lithium wrong in exactly the way every other calculation gets lithium wrong. If the chain had gotten lithium right using the same equations, that would be suspicious — it would indicate a hidden parameter or an error that accidentally compensated. Getting it wrong the same way confirms the chain is running standard physics. The lithium problem is real. It's in the nuclear physics, not in the parameterization. The chain located the problem precisely by reproducing it.

A separate function quantifies the tension:

```

def bridge_li7_problem_v0(value_dicts):

```

```

vm = _value_map(value_dicts)

li7_derived = mpf(str(_get(vm, "result_li7_derived_v0")))

li7_measured = _mpf_val(vm, "cosmo_li7_measured_v0")

ratio = li7_derived / li7_measured

is_real = ratio > mpf("2")

li7_a = _mpf_val(vm, "bbn_li7_a_coeff_v0")

li7_b = _mpf_val(vm, "bbn_li7_b_coeff_v0")

li7_meas_x1e10 = li7_measured * mpf("1e10")

eta10_needed = mpf("6") + (li7_meas_x1e10 - li7_a) / li7_b

eta10_derived = mpf(str(_get(vm, "result_eta10_derived_v0")))

return {
    "key": "bridge_li7_problem_v0",
    "outputs": {
        "result_li7_problem_ratio_v0": _approx(ratio),
        "result_li7_problem_is_real_v0": is_real,
        "result_eta10_needed_for_li7_v0": _approx(eta10_needed),
        "result_eta10_derived_v0": _approx(eta10_derived),
        "result_eta10_li7_tension_v0":
            _approx(abs(eta10_derived - eta10_needed)),
    },
}

```

Seven lines of arithmetic. The ratio of predicted to observed lithium. A boolean: is the ratio greater than 2? The inverse question: what η_{10} would lithium need to match? Answer: approximately 1.4, versus the derived 6.09. The tension quantified in η_{10} units.

The lithium problem that has been discussed in hundreds of papers across decades of research, located in seven lines of Python by an outsider on a laptop.

5. The Value Store

Every input to every computation exists as a named, typed, versioned entry with full provenance. The store contains over 5,000 entries. Each entry carries a unique key, the value itself stored as an exact fraction where possible, the unit, the number of significant digits, the uncertainty, the source, tags for searchability, and notes for methodological context.

A representative entry:

```

NODE: si_boltzmann_constant_v0

canonical      si_boltzmann_constant

key            si_boltzmann_constant_v0

level          0

node_type      value

notes          Exact since 2019 SI revision. Defines the kelvin.

source         SI 2019. Exact by definition. k_B = 1.380649e-
23 J/K.

tags           ["SI", "exact", "thermodynamic"]

term           boltzmann_constant

topic          si

unit           J/K

value          {"num": "1380649", "den": "10^29"}

value_type     exact_fraction

version        0
```

Every field serves a purpose. The key enables machine reference — no ambiguity about which value is being used. The source enables tracing — anyone can verify against the

original publication. The value type states whether the representation is exact or approximate. The notes capture context at the point of data entry.

The BBN experiment draws from twenty-eight named entries. Each is verifiable against its cited source. The runner loads only declared entries. An undeclared value cannot enter the computation. If a function references a key that was not declared in the experiment definition, the runner fails with a traceable error. The anti-smuggling property is architectural — the infrastructure has no mechanism for an undeclared value to participate in a derivation.

The fraction list — values that are already exact fractions before any 2^{335} work — demonstrates what the field already has and does not use:

Beta coefficients: 41/10, -19/6, -7, 25/6, -13/6, -20/3. Group theory: 3/4, 4/3, 2, 3, -11/3. Representation theory: 1/15, 1/3, 1/6, 2/3, 1/2, 4/3. SI constants: exact integers and fractions by definition. QED analytical terms: 1/2, 197/144, 3/4, 1/12, 28259/5184. Gauge structure: 22/13, 38/27, 44/169, 218/115. CKM matrix elements as rationals with explicit precision: 22501/100000, 737/200000, 2091/50000.

Every one of these is from the field's own published results. Every one is an exact fraction. Every one is available in any textbook or reference compilation. The field derived them. The field published them. The field does not compute with them in their exact form. The value store does.

6. The Run

```
$ ./data7.py run experiment_bridge_bbn_v0
```

```
DATA-6 RUNNER: experiment_bridge_bbn_v0
```

```
Source: experiment_bridge_bbn_v0.json
```

```
Mode:    standard
```

```
Purpose: program_parameter_reduction_v0
```

```
Loaded 4904 value nodes.
```

```
EXECUTION PLAN: 7 derivations
```

```
[OK] bridge_omega_b_from_integers_v0      8 outputs
```

```
[OK] bridge_omega_de_from_flatness_v0     8 outputs
```

```
[OK] bridge_eta_from_omega_b_v0          9 outputs
```

[OK] bridge_yp_from_eta_v0	7 outputs
[OK] bridge_dh_from_eta_v0	9 outputs
[OK] bridge_neff_consistency_v0	8 outputs
[OK] bridge_vacuum_energy_v0	10 outputs

Derivations: 7 OK, 0 errors

COMPARISONS: 13 checks

[INFO] Omega_b from integers vs Planck
predicted 0.049036 ref 0.0490 miss 0.073%

[INFO] eta derived vs Planck
predicted 6.090e-10 ref 6.104e-10 miss 0.237%

[INFO] Y_p from integers vs measured
predicted 0.2486 ref 0.2449 miss 1.528%

[FAIL] Y_p digits of agreement
expected 0.245 got 0.249 miss 1.487%

[INFO] D/H from integers vs measured
predicted 2.531e-5 ref 2.527e-5 miss 0.143%

[PASS] D/H digits of agreement
3-digit match, miss 0.024%

[PASS] N_eff consistency with N_eff = 3
got 2.712 range [2.5, 3.5]

[PASS] Full chain: integers → Y_p within 1 sigma
got 0.936σ range [0, 2.0]

[PASS] Full chain: integers → D/H within 1 sigma

got 0.120σ range [0, 2.0]

EXPERIMENT SUMMARY

Derivations: 7 / 7

PASS: 4 FAIL: 1 INFO: 8 SKIP: 0

STATUS: 1 FAILURES

Seven derivations. Zero errors. Fifty-seven output values. Thirteen pre-registered comparisons with mechanical verdicts. Four passes. One failure — Y_p digit agreement, printed with the same precision as the passes. Eight informational results with quantified misses. Every output traceable to a specific run. Every input traceable to a specific source publication.

The output is deterministic. Run it today. Run it in ten years. Same fifty-seven numbers. The reference values may shift — CODATA will reprocess, Planck will be superseded by future surveys. The derived values will not shift because exact arithmetic does not drift. The computation is more stable than the references it is compared against.

7. What the Result Means

The derivation chain connects gauge integers at the Standard Model scale to nuclear abundances at the Big Bang Nucleosynthesis scale through cosmological parameters measured by satellite. It spans three physics domains that are practiced by three separate communities, published in three separate journal families, funded by three separate grant programs, and taught in three separate course sequences.

The chain has not been published by any institutional physicist. Not because it is hard. The code in Section 4 demonstrates that it is trivial. Each function is a textbook equation translated into Python. Each is readable by a first-year CS student. Each is writable by a first-year CS student given the equation and the value store keys.

The chain has not been published because the institutional structure of physics has no home for it. The particle physics department does not derive nuclear abundances. That is the cosmology department's domain. The cosmology department does not start from gauge integers. Those belong to the particle physics department. The nuclear physics department does not connect to gauge structure. That is high-energy theory's domain. Each department works within its boundary. The chain crosses all three boundaries. No department owns it. No journal specializes in it. No conference program has a session for it. No career track rewards it.

An outsider has no departmental boundary to cross. The chain is one computation. The three domains are three sets of equations. The equations are all published. The values are all available. The connection is arithmetic. The departmental structure that prevented

the chain from being assembled is an institutional artifact. The universe does not have departments.

8. The Hierarchy Inversion

Physics has two outputs that other fields need: equations and measurements. The equations describe how physical quantities relate to each other. The measurements provide the numerical values of those quantities. Both are essential. Both are the product of centuries of brilliant work by thousands of dedicated scientists.

Both are now commodity inputs.

The equations are published in textbooks, papers, and reference compilations. They are available to any LLM user. The Standard Model Lagrangian, the BBN fitting formulae, the QED perturbative series, the cosmological parameter relations — all accessible by asking a question. The LLM has been trained on the physics literature. It has read more physics papers than any living physicist. This is not a comment on any physicist's capability. It is a comment on the volume of literature. No human reads the entire corpus. The LLM was trained on it.

The measurements are published in CODATA compilations, PDG reviews, Planck data releases, and specialized surveys. They are available as numbers in papers that anyone can access.

The work that remains — the work that produces new results — is the engineering work of connecting equations across domains, implementing them in exact arithmetic, tracking provenance on every input, pre-registering comparisons, building reproducible infrastructure, and producing auditable cross-domain results. This is software engineering. Data engineering. Applied mathematics. Systems architecture. Quality assurance. Configuration management.

These are not physics skills. They are not taught in physics departments. They are standard practice in software engineering and are absent from physics.

The physicist was trained for a decade to learn the equations. The outsider asked an LLM. The physicist performs computation in floating-point without version control, without input validation, without provenance tracking, without automated testing. The software engineer does all of these as standard practice. The physicist writes a paper describing the computation in prose. The software engineer publishes the computation as executable code with declared inputs and automated verification.

The skills that modern physics requires — cross-domain breadth, exact arithmetic, provenance tracking, pre-registered testing, reproducible infrastructure — are engineering skills. The skills that physics departments teach — equation derivation, perturbative methods, field theory formalism — are available through LLMs. The skills that are scarce are not the physics skills. The skills that are scarce are the engineering skills.

The hierarchy has inverted. Physics provides equations and measurements — essential but commodity. Engineering provides the infrastructure to connect them, track them, verify them, and produce auditable cross-domain results. The institution has not recognized the inversion because recognizing it would mean acknowledging that the credential gate is defending a knowledge monopoly that no longer exists.

The modern physics experiment is a sensor array producing gigabytes of time-series data processed by code. The physicist’s role is increasingly to write analysis code. The physicist writes analysis code without the training, tools, or discipline that software engineers apply to code that must work reliably. The institution that claims to produce the most precise knowledge in science does not apply basic quality assurance to its computational methods.

The physics community claims to be searching for unification. Unification requires cross-domain derivation. Cross-domain derivation requires breadth across departments plus engineering discipline. The credential gate excludes the people with the engineering discipline. The departmental structure prevents the breadth. The floating-point arithmetic introduces noise that exact fractions would eliminate. The prose-based publication format prevents reproducibility. Every structural feature of the institution works against the goal the institution claims to pursue.

9. No Path

The work described in this paper exists. The code is public. The value store is public. The experiment definitions are public. The results are public. The DOIs are registered. The timestamps are recorded. Everything needed to verify the work is available to anyone with Python installed.

The work has no path into the field’s evaluation process.

arXiv. The open preprint server for physics. Account creation requires institutional affiliation or endorsement from an existing arXiv author. The author of this work attempted to create an account and could not. The gate is at account creation — before submission, before review, before content evaluation. The most permissive endpoint in the field’s communication infrastructure rejected the work before it could be entered into the system.

Journal submission. Requires credentials the author does not have. Major physics journals perform editorial screening that includes assessment of author affiliation and publication history. A submission from an unaffiliated author claiming that gauge integers predict nuclear abundances would be rejected at editorial screening. Content evaluation — peer review — occurs after editorial screening. The content is never reached.

Conference presentation. Abstract selection committees apply credential filters. An abstract from an unaffiliated author would be classified based on the pattern — outsider claims fundamental result — before the abstract’s content was evaluated.

Collaboration. Requires an institutional insider willing to stake professional reputation on outsider work. Requires the insider to have encountered the work. Requires the work to be on a platform the insider monitors. Requires access to that platform. The circularity

is complete: you need an insider to get evaluated, you need to be evaluated to be taken seriously, you need to be taken seriously to attract an insider.

The work is published on Zenodo — a legitimate open-access repository maintained by CERN, with DOI registration, version control, and permanent archiving. The code is on GitHub. A book has been written making the methodology accessible. A series of papers — the HOWL-CULT series — documents the structural problems this case study illustrates. Every artifact is public, timestamped, and auditable.

The physics community does not monitor Zenodo for outsider contributions. It does not monitor GitHub for physics value stores. It does not read books by unaffiliated authors about fundamental physics. The work is published in the literal sense — it is public, permanent, and available. It is unpublished in the institutional sense — it does not exist within the system the field recognizes as publication.

The content is never evaluated. Not because it was examined and found wanting. Not because someone ran the code and got different results. Not because someone checked the value store and found errors. Not because someone identified a logical gap or a mathematical mistake or an empirical disconnect. The content is never evaluated because every path to evaluation requires passing a credential filter, and the credential filter is the first gate on every path.

10. The LLM Discontinuity

The credential gate was a knowledge gate. Before large language models, physics knowledge was locked in technical literature, specialized courses, and institutional training programs. Accessing the full body of knowledge required years of formal education. The credential was a proxy for knowledge. The proxy was imperfect but not unreasonable because the knowledge was genuinely inaccessible without the training pipeline.

That era is over.

Any person with access to an LLM can retrieve any published equation from any subfield of physics. Can retrieve any measured value from any reference compilation. Can retrieve derivation methods, fitting formulae, perturbative series structures, group theory results, and cosmological parameters. Can get working code that implements any of these. Can get explanations of the methodology at any level of detail.

The LLM does not replace the physicist's judgment. It does not replace experimental design. It does not replace the intuition that comes from years of working with specific physical systems. What it replaces is the knowledge barrier that the credential gate was designed to enforce.

The credential gate now filters on social membership, not on knowledge. The knowledge is outside the gate. The tools are outside the gate. The engineering skills are outside the gate. The results — as demonstrated by the BBN cross-domain chain — are outside the gate. The only thing inside the gate is the social structure that determines whose work gets evaluated.

The pool of people who can do valid cross-domain physics has expanded from a small community of credentialed specialists to anyone with an LLM, engineering discipline, and the willingness to declare inputs, pre-register comparisons, and print failures. This is not a future prediction. It is a present fact demonstrated by the existence of the work described in this paper.

The institution can adapt by implementing content-based evaluation — evaluating work on its logic, its empirical grounding, its mathematical correctness, and its utility, regardless of who produced it. A framework for such evaluation is described in [HOWL-CULT-14-2026].

Or the institution can maintain the credential gate and watch as valid work accumulates outside the institution, verified by anyone who runs the code, producing results the institution has not produced, using the institution’s own equations and values. The work will not stop because the institution refuses to look at it. The LLMs will not stop providing access to the knowledge. The exact arithmetic will not stop producing stable results. The value stores will not stop growing. The cross-domain chains will not stop landing.

11. The Resolution

This case forces the resolution that [HOWL-CULT-14-2026] demanded. The institution has three options.

Option 1: Evaluate outsider work on content. Implement a mechanical content-based evaluation process. Every submission, regardless of author credentials, is evaluated against defined criteria — logical consistency, empirical grounding, mathematical correctness, utility. The evaluation produces a located result that can be explained to the submitter with specific reference to their work. If the BBN chain fails content evaluation, the failure is located at a specific step, and the submitter knows what to fix. If it passes, the field gains a result it did not have.

Option 2: Declare the gate closed. State publicly, formally, and without ambiguity that submissions from authors without specified credentials are not accepted. No one wastes time building work that will never be examined. The institution drops its claim to content-based evaluation and accepts the consequences for its credibility.

Option 3: Continue the ambiguity. Nominal openness with practiced exclusion. This option was defensible before this case because the cost was invisible — the institution could not know what it discarded because it never examined what it discarded. This case makes the cost visible. A cross-domain derivation chain using the field’s own equations and values, producing results the field has not produced, fully auditable, fully reproducible, sitting in public repositories with DOIs. The cost of the credential gate is no longer invisible. It is documented, timestamped, and reproducible.

Before this case, the institution could plausibly claim that the credential gate’s false rejection rate for scientifically valid work was zero or negligibly small. This case eliminates that claim. The false rejection rate is nonzero. The rejected work is specific, auditable,

and reproducible. The rejection occurred at account creation on the preprint server. The content was never evaluated.

One case is sufficient to force the resolution because one case is sufficient to demonstrate nonzero false rejection. The gate’s defenders argue that it filters noise efficiently and the cost of occasional false rejection is acceptable. That argument requires the false rejections to be invisible. This one is not. It has a DOI. It has a GitHub repository. It has running code. It has fifty experiments. It has five thousand tracked values. It is sitting in public, verified, documented, and ignored.

12. Falsification Conditions

This paper commits to specific conditions under which its claims would be false.

Claim 1: The BBN cross-domain chain is mathematically valid. If the chain contains a mathematical error at a specific step — an incorrect equation, a misapplied formula, a wrong value — the chain fails at that step. The code is public. Anyone can check every line. Anyone can run the code and verify every output. If the results do not reproduce, the chain is invalid. The command is `./data7.py run experiment_bridge_bbn_v0`.

Claim 2: No institutional physicist has published this chain. If an institutional publication exists that derives primordial nuclear abundances from gauge integers through a single declared derivation chain with pre-registered falsification conditions, this claim is false. The author has searched the literature with LLM assistance and found no such publication. The search may have missed one. If so, the omission should be identified and the claim corrected.

Claim 3: The credential gate prevented evaluation. If a path exists for an unaffiliated author with no physics publication history to submit work for content evaluation at a major physics journal or preprint server without credential filtering, this claim is false. The author’s experience at arXiv is documented. Alternative paths may exist that the author did not discover. If so, they should be identified.

Claim 4: The code is trivial. If the derivation functions require physics knowledge beyond what an LLM provides, or programming skill beyond first-year CS level, the triviality claim is false. The code is published. Its complexity can be assessed by any reader. If the functions require specialized knowledge or advanced programming technique, the claim fails.

Claim 5: The hierarchy has inverted. If the cross-domain result could have been produced with standard physics tools and practices — floating-point arithmetic, prose-described computation, no value store, no pre-registered comparisons — without the engineering infrastructure described in this paper, then the engineering skills were not the differentiating factor and the hierarchy inversion claim is weakened. The test: reproduce the BBN chain using standard physics computational practice and compare the auditability, reproducibility, and provenance of the result.

Each condition is specific. Each is observable. Each could fail. The paper commits and accepts falsification if it comes.

13. Closing

The command is:

```
./data7.py run experiment_bridge_bbn_v0
```

The output is fifty-seven values. The comparison is thirteen tests. The verdict is mechanical. Four passes. One failure. Eight informational. Every input sourced. Every step auditable. Every failure printed.

The equations are from textbooks. The values are from CODATA and published measurements. The arithmetic is exact. The code is Python that a first-year CS student can read, and can write. The result crosses three physics domains that no institutional publication has connected in a single derivation chain. The time investment was three months. The physics background was high school level. The engineering background was decades of software development. The LLM provided the physics knowledge. The engineer provided the discipline.

The work is published. It is on Zenodo with DOIs. It is on GitHub with full source. It is documented in a series of papers with cross-references. A book has been written. Every artifact is public, timestamped, and auditable.

The field cannot evaluate it. Not because the field has examined it and found it wanting. Because the field's evaluation process requires credentials the author does not have, starting at account creation on the preprint server. The content has never been examined by anyone inside the institution.

The institution claims to search for unification. Unification requires cross-domain work. Cross-domain work has been done. It is sitting in public. It uses the institution's own equations and values. It produces results the institution has not produced. The institution will not look at it because of who did it, not because of what it contains.

The physics is trivial. The engineering is standard. The result is real. The path is closed.

The universe doesn't check credentials. It evaluates interactions on their content — energy, momentum, spin, charge. The outcome is determined by what the interaction contains, not by where the participants came from. A scientific institution studying this universe could adopt the same principle.

Run it.

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Series cross-references (for deeper treatment of concepts introduced in this paper):

- Statistical breadth substituting for functional depth: [\[HOWL-CULT-1-2026\]](#)
- The institutional gap between measurement domains: [\[HOWL-CULT-2-2026\]](#)
- The structural cessation of falsification practice: [\[HOWL-CULT-3-2026\]](#)
- What falsification structurally requires: [\[HOWL-CULT-4-2026\]](#)
- Why located errors are the most valuable finding: [\[HOWL-CULT-5-2026\]](#)
- Publications as immutable timestamps: [\[HOWL-CULT-6-2026\]](#)
- Inherited enforcement of untested norms: [\[HOWL-CULT-7-2026\]](#)
- Working in the space between departments: [\[HOWL-CULT-8-2026\]](#)
- Institutional structure for cross-domain work: [\[HOWL-CULT-9-2026\]](#)
- Closing domains as a repeatable method: [\[HOWL-INFO-10-2026\]](#)
- The structural mechanics of institutional non-commitment: [\[HOWL-CULT-12-2026\]](#)
- Replacing averaged snapshots with synchronized measurement: [\[HOWL-CULT-13-2026\]](#)

- Content-based evaluation of scientific contributions: [\[HOWL-CULT-14-2026\]](#)
- Structural specification for reproducible computation: [\[HOWL-CULT-15-2026\]](#)
- Why the physics community will never close its open problems: [\[HOWL-CULT-16-2026\]](#)

[\[HOWL-CULT-17-2026\]](#) No Path: Cross-Domain Derivation by an Outsider Using Commodity Tools. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-17-2026>

[\[HOWL-CULT-1-2026\]](#) The N=1000 Fallacy. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-1-2026>

[\[HOWL-CULT-2-2026\]](#) The Domain Boundary. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-2-2026>

[\[HOWL-CULT-3-2026\]](#) The Falsification Forfeiture. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-3-2026>

[\[HOWL-CULT-4-2026\]](#) What Falsification Actually Requires. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-4-2026>

[\[HOWL-CULT-5-2026\]](#) Falsification as Finding. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-5-2026>

[\[HOWL-CULT-6-2026\]](#) Immutable and Mortal. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-6-2026>

[\[HOWL-CULT-7-2026\]](#) The Five Monkeys Problem. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-7-2026>

[\[HOWL-CULT-8-2026\]](#) The Omni-Domain Investigator. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-8-2026>

[\[HOWL-CULT-9-2026\]](#) The Omni-Academy. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-9-2026>

[\[HOWL-CULT-12-2026\]](#) Cowardice and High Status as Anti-Popper Progress Blocking. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-12-2026>

[\[HOWL-CULT-13-2026\]](#) Scientific Time Series Data: A Proposal for a 21st Century Upgrade to CODATA. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-13-2026>

[\[HOWL-CULT-14-2026\]](#) Mechanical Evaluation of Outsider Scientific Contributions.

Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-14-2026>

[[HOWL-DATA-6-2026](#)] A Versioned Node System for Integer Fraction Physics. Github: <https://github.com/ghowland/papers/tree/main/papers/DATA/HOWL-DATA-6-2026>

[[HOWL-CULT-15-2026](#)] Showing Your Work in 2026. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-15-2026>

[[HOWL-CULT-16-2026](#)] It's Hard: Why the Physics Community Will Never Close Its Open Problems. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-16-2026>

[[HOWL-INFO-10-2026](#)] Closing Domains. Github: <https://github.com/ghowland/papers/tree/main/papers/INFO/HOWL-INFO-10-2026>